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Physics-Informed Surrogate Neural Network for Optimizing Diode-Based Interfaces in Leadless Pacemaker Energy Harvesting

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***Abstract*—Efficient AC–DC interfaces are essential for low-power energy harvesters to power intracardiac leadless pacemakers (ICLPs) reliably. For the front-end circuitry, selecting a diode that minimizes both conduction and leakage losses under specified operating conditions is challenging, as the forward voltage drop and reverse leakage current cannot be simultaneously reduced due to the device's inherent physics. This brief presents a machine learning framework that predicts the DC output of diode-based AC-DC interfaces directly from specific voltage-current (V-I) characteristic points of the diodes and descriptive features of the excitation waveform. A physics-informed surrogate model is trained on SPICE-simulated data of rectifiers driven by experimentally captured cardiac harvester waveforms. Hardware validation with printed-circuit rectifiers powered by the same harvester source exhibits an average error of 0.03 V between prediction and measurement, matching the performance observed on the synthetic set and demonstrating robust generalization to real-world applications. This data-driven approach offers an instant, datasheet-only diode screening without circuit-level evaluation and can be integrated into automated multi-objective optimization flows for the design of low-power energy harvester AC–DC interfaces.**

***Index Terms*—AC–DC Interface, Diode Screening, Energy Harvester, Machine Learning, Physics-Informed Neural Network**

I. Introduction

THE rapid development of wearable, medical implants, and distributed Internet of Things (IoT) devices has intensified research in miniature energy harvesters. Piezoelectric energy harvesters (PEH) enable long-term power for intracardiac leadless pacemakers (ICLPs), removing the need for battery-replacement surgeries, as the heart provides kinetic energy [1]. A contact-based impact piezoelectric method, validated in animal trials, generates pulsating AC power in the tens of microwatts range [2], with short pulse durations due to shock-driven heart motion [3]. Efficient use of short-duration, low-power energy for ICLPs requires a high-performance and miniature AC–DC interface to rectify the harvested power and deliver it to storage [4].

Most interfacing circuits rely on diodes as switches to realize AC-DC conversion. However, diode non-idealities impose significant penalties. The forward voltage drop dissipates a proportion of the harvested power, reverse leakage current drains the storage element, and finite reverse-recovery time adds loss at every polarity reversal [5]. Bridgeless or synchronous interfaces promise higher efficiency, but at microwatt levels, the control overhead often cancels the gain [6], [7]. Besides, the complex circuit architecture enlarges the overall size and exceeds the limitations of ICLP systems, where all components must be mounted in a cylinder measuring 6 mm in diameter and 2 mm in height [2]. Consequently, traditional diode-based structures remain practical choices for low-power harvesters. Within such frameworks, selecting appropriate diodes whose electrical properties carefully match specific operating conditions is essential for maximizing efficiency and utilizing the scarce harvested energy [8].

In steady state, the diode voltage-current (V-I) characteristic can be approximated by the Shockley equation: $I_D = I_S\left(e^{V_D/nV_T} - 1\right)$ [9], where I_D and V_D denote the diode current and voltage, respectively. I_S is the diode's reverse-bias saturation current, n is the ideality factor, and V_T is the thermal voltage. According to the equation, for a specified forward conduction current, achieving a lower forward voltage drop across the diode requires a larger saturation current I_S, which magnifies the leakage current under reverse bias, and vice versa. This intrinsic trade-off complicates device selection for rectifier optimization.

For a given operating condition defined by the harvester's properties, an optimal diode exists that minimizes the combined forward conduction and reverse leakage losses, as lowering the forward voltage drop invariably increases the leakage current and vice versa. In practice, however, identifying an appropriate diode is far from straightforward, because rectifier efficiency depends strongly on both device properties and operating conditions [10], and data-sheet parameters alone rarely predict in-circuit performance.

Information from the diode datasheet, including V-I characteristic plots, cannot comprehensively evaluate the interplay between forward and reverse losses. Since miniature harvesters operating at low power density and low duty cycle can impose different requirements on diode properties, how they can match each other is hard to estimate [8]. Thus, designers struggle to perform a fair, quantitative comparison of candidate devices. In practice, engineers rely on experience to pick diode candidates, followed by conducting iterative simulations or prototype tests. Each screening loop relies on lengthy transient simulations with recorded harvester waveforms or laboratory measurements, for example, a 120-s trace sampled at 5 kHz needs over 60 s to run, severely limiting rapid diode evaluation. Furthermore, comparative simulations require validated SPICE models rather than simple datasheet information.

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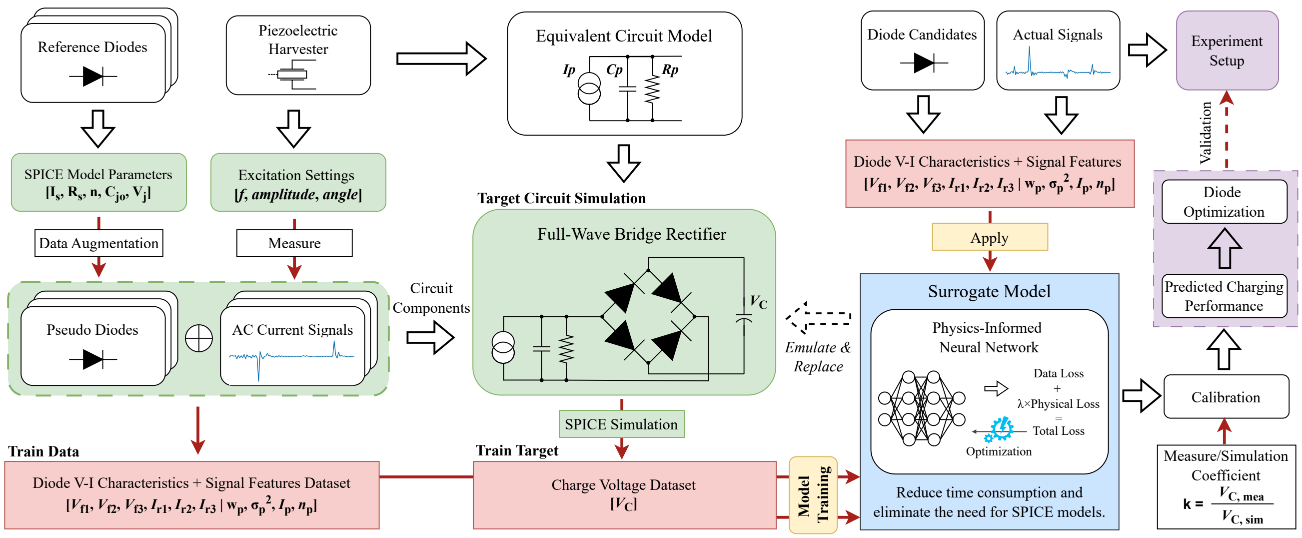

Fig. 1. Overview of the PINN-assisted diode screening framework

For diodes without such models, simulation is not applicable unless manual parameter extraction is performed [11], which may introduce extra uncertainty and workload. Lacking a direct analytical link between datasheet parameters and in-circuit performance, this empirical workflow is slow and risks overlooking better diodes.

To overcome these challenges, we are motivated by machine learning (ML) techniques that are enhancing low-power electronics [12] and automated circuit synthesis [13], [14]. Datasheet-driven ML surrogates have achieved fast inductor characterization for automated filter design [15], and physics-informed neural networks (PINN) have been employed for converter parameter extraction from measured waveforms [16].

In this paper, we introduce a data-driven framework to evaluate diode candidates for the AC-DC rectifier circuit through an ML surrogate model that predicts conversion performance directly from a set of diode V-I characteristic points and statistical features of the power waveform. We implement the model as a PINN [17], trained on synthetic data generated by SPICE rectifier simulation with experimentally recorded cardiac PEH power waveforms [2]. Furthermore, practical measurements on rectifiers built with unseen diodes reveal only a minimal average discrepancy between predicted and measured outputs, which aligns well with the results on the synthetic data and confirms robust generalization. By eliminating repeated prototype iterations and exhaustive simulations with SPICE models, the proposed tool accelerates diode screening to maximize energy-conversion efficiency for PEHs, which enables leadless pacemakers and offers a possibility to realize automated multi-objective optimization flows for interface design under given operating conditions.

II. Methods

A. Framework Overview

Recent advances in ML regression have produced surrogate models that can approximate circuit simulation results with negligible computation time [14], [18]. This capability enables a direct mapping from two readily available inputs: (i) datasheet V-I characteristics points of the diode and (ii) descriptive features of the energy harvester power waveform, to the conversion performance metric. Fig. 1 summarizes the workflow of the proposed framework, which comprises synthetic data generation from SPICE simulation, surrogate model construction, and experiment validation.

We demonstrate the framework on a full-wave bridge rectifier that conditions the output of a miniature implantable piezoelectric harvester [2]. Since application constraints limit the overall size and preclude complex interfacing topologies, the conversion efficiency largely depends on the choice of a diode that offers the optimal trade-off between forward voltage drop and reverse leakage.

The overall conversion performance of the AC-DC rectifier is quantified by the maximum voltage V_C on the output storage capacitor at steady state, closely mimicking charging a micro-battery in practical systems. This metric can reflect the combined influence of the forward conduction loss during charging intervals and the dissipation due to reverse leakage during discharge intervals, as the input power equals the sum of these losses at steady state.

B. Synthetic Dataset Generation

Training samples comprise 7200 input-output pairs. The inputs combine diode and waveform features. The output labels are steady-state capacitor voltages V_C, simulated from the rectifier driven by experimental cardiac harvester waveforms.

We start from 75 commercial small-signal switching diode SPICE models for their short reverse recovery time and small size suited to low-power operation. This baseline is augmented as 300 pseudo-diodes by independently perturbing their original SPICE parameters (saturation current I_S, ideality factor n, series resistance R_S, zero-bias junction capacitance C_{j0}, and junction potential V_j [19]) within ± 5 %, preserving realistic correlations.

The variant current waveforms for simulation are produced by an electrodynamic shaker that generates cardiac vibration

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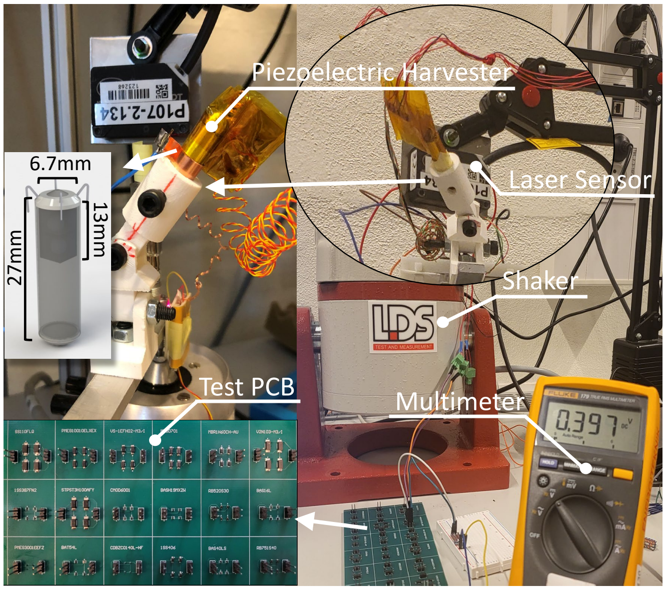

Fig. 2. Experiment Setup

profiles and excites the PEH [2]. The PEH current pulses are recorded as the simulation stimulus by measuring the short-circuit current. To ensure the waveform diversity, 27 sets of current waveforms are collected at shaker frequencies of 1.0, 1.2, and 1.4 Hz; input cardiac signal amplitudes of 600, 650, and 700 mV; and tilt angles from 0° to 45°. From every sample, four descriptive statistics are extracted, including the mean width of pulses w_p, the variance of pulse width σ_p^2, the mean amplitude of pulses I_p, and the mean number of pulses per minute n_p, which are appended to the inputs as the waveform features.

The AC-DC interface circuit under study is a full-wave bridge comprising four identical diodes that charge a 10 μF output capacitor in parallel with a 5 MΩ insulation resistor, representing the self-leakage. The piezoelectric harvester is modeled by its simplified equivalent model: a current source shunted by its intrinsic capacitance C_p and internal resistance R_p [20]. The complete SPICE netlist is assembled by placing the pseudo-diode model as the rectifier components and applying one of the recorded waveforms as the source current. A 120 s transient simulation is then executed to reach steady state, and the maximum capacitor voltage V_C within the final 10 s is sampled as the target label. Iterating over all diode and waveform combinations yields the synthetic dataset.

C. PINN Surrogate Model

Within the proposed framework, PINN [17] acts as the non-parametric surrogate model that maps the input features to the steady-state capacitor voltage V_C to evaluate diode performance in the target conversion circuit. Prior to training, all data are preprocessed by standardization.

The PINN adopts the deep neural network (DNN) architecture with a physics penalty based on power balance at steady state. It uses a fully connected 5-layer network with hidden widths of 64, 64, 32, 16, and 2 and activated by tanh, with a dropout rate of 0.1 and a batch size of 400. Inputs are standardized and the training is carried out over 700 epochs using the Adam optimizer. These parameters are determined via an exhaustive grid search where architectures with 3 to 6 hidden layers, widths from 2 to 256, and activation functions of ReLU, Swish, and tanh are explored. In this framework, the physical constraint is the condition for V_C to reach the steady-state balance point, at which the input power P_{in} is equal to the total loss P_{loss}.

Given a predicted voltage on the capacitor $V_{C,\,\text{pred}}$, the input power can be computed by the capacitor voltage times the input current, which can be given by:

$$P_{\text{in}} = \frac{1}{2} V_{C,\,\text{pred}} \left(I_p - V_{\text{piezo}}/R_p \right) \frac{1}{60}\, w_p\, n_p, \tag{1}$$

where w_p, n_p, and I_p are the waveform statistics included in the input features. The piezoelectric source voltage is estimated as $V_{\text{piezo}} = V_{C,\,\text{pred}} + 2\,V_{\text{fwd}}$ and thus V_{piezo}/R_p yields the internal loss current of the harvester. The forward voltage drop V_{fwd} at current I_p is calculated by the Shockley equation (1), whose coefficients are fitted from the input V-I characteristic pairs of the target diode.

Furthermore, the loss part consists of the diode reverse leakage and capacitor self-discharge:

$$P_{\text{loss}} = I_{\text{leak}}\, V_{C,\,\text{pred}} + \frac{V_{C,\,\text{pred}}^2}{R_C}, \tag{2}$$

where R_C is the insulation resistance of the capacitor. The current leakage I_{leak} is approximated by an empirical exponential model fitted to the input diode V-I features.

The composite loss for model training is given by:

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{\text{data}} + \lambda\, \mathcal{L}_{\text{phys}} \\ &= \frac{1}{N} \sum_{i=1}^{N} \Big[\left(V_{C,\,\text{pred}} - V_{C,\,\text{true}} \right)^2 + \lambda \left| P_{\text{in}} - P_{\text{loss}} \right| \Big], \end{aligned} \tag{3}$$

where the first term is the mean squared error (MSE) data loss, and the second is the mean absolute error (MAE) for physical loss, weighted by a scalar coefficient λ.

D. Calibration

To compensate the systematic bias between simulated and measured peak capacitor voltages for more accurate prediction, we compute a scalar calibration factor $k = \overline{V}_{C,\,\text{mea}}/\overline{V}_{C,\,\text{sim}}$ from an independent set of calibration diodes and scale all predictions as $V_{C,\,\text{calib}} = kV_{C,\,\text{pred}}$, where $\overline{V}_{C,\,\text{mea}}$ and $\overline{V}_{C,\,\text{sim}}$ are the average measured and simulated charge voltages, respectively. Since systematic errors may vary with operating conditions, calibration factors should be determined separately for each excitation or measurement setup.

III. Experiment Setup

A B&K V406 shaker mimics cardiac motion to excite the PEH, producing the current waveforms for synthetic data generation and for driving practical circuits, with laser sensing ensuring accurate motion as presented in Fig. 2. Details on the PEH and motion simulation are available in [2]. For experimental validation, 13 test diodes (10 without validated SPICE models and cannot be simulated) and 6 calibration diodes are mounted on a printed circuit board (PCB).

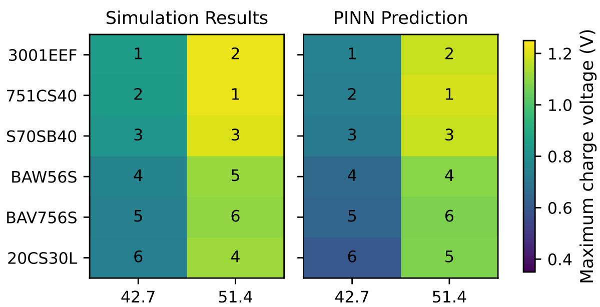

Fig. 3. Heat map and diode relative rankings of PINN prediction and simulation (numbers indicate diode rankings)

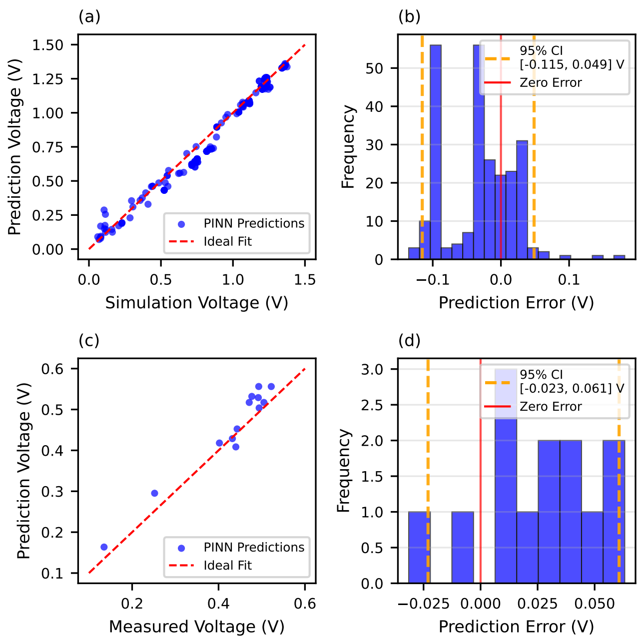

Fig. 4. (a) PINN prediction vs simulation voltage, (b) PINN prediction vs simulation error histogram, (c) PINN prediction vs measurement voltage, (d) PINN prediction vs measurement error histogram

IV. Results

The proposed framework is coded in Python, circuit simulations are performed with PySpice, and the PINN surrogate model is implemented in TensorFlow. The test set comprises 900 samples simulated from the original diodes under a 650 mV, 1.2 Hz stimulus, while the training set consists of 7200 samples (5% as a validation set) from pseudo-diodes with any other amplitude–frequency combinations. The PINN achieves notable computational speed, requiring around 270 s for model training and 47.5 ms for single sample prediction compared to 60-70 s for SPICE simulation (120 s trace at 5 kHz). Furthermore, regression accuracy is evaluated by mean absolute error (MAE), root mean squared error (RMSE), Pearson correlation coefficient ρ, and p-value from paired t-tests.

A. Prediction vs Simulation

Fig. 3 compares the peak capacitor voltages V_C predicted by the PINN model with their simulation values for six representative diodes under three excitation levels. Each heat map cell is annotated with the diode performance ranking under the corresponding operating conditions. Although the diode ranking varies with excitation conditions, the PINN can reproduce the correct order in almost every case. The few mismatches occur mainly when the simulated voltages differ by less than 50 mV.

The performance of the PINN surrogate is presented in Fig. 4 (a), which plots the predicted versus simulated V_C for the entire test set. The points tightly follow the unity-slope line, indicating consistent regression within the test range. As reported in Table I, the model attains a mean absolute error of 0.047 V and a Pearson correlation coefficient of 0.989 on the test data. Despite slightly greater dispersion in predicting lower voltages, PINN can estimate the diode performance from its characteristics and features of the input waveform with sufficient accuracy to replace circuit-level simulation for component evaluation.

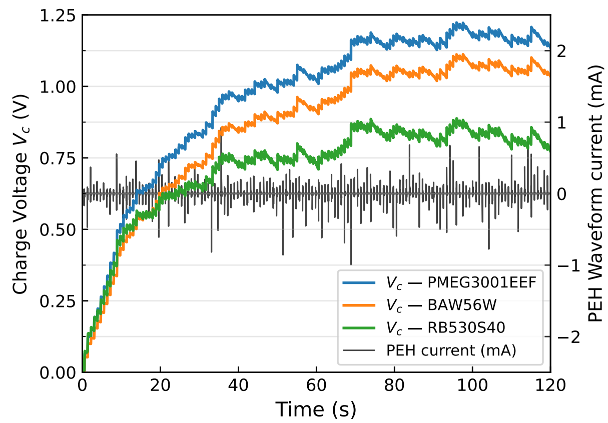

Fig. 5. Comparison of diode conversion performance

Fig. 5 illustrates the simulation charged voltage over time of the optimal diode chosen by the proposed framework compared with two representative diodes. The appropriate diode contributes to boosting the charged voltage, demonstrating the benefit of the diode selection on the conversion efficiency.

B. Prediction vs Measurements

To assess generalization, PINN predictions are validated against measurements from 13 PCB rectifiers using unseen commercial diodes and harvester waveforms not present during model training, replicating real-world applications. Fig. 4 (b) plots the predicted V_C after calibration ($k = 1.07$ derived from calibration) versus the measured voltages. The plot demonstrates that the PINN model achieves similar performance on both the measurement and synthetic sets. The measurement-prediction points are clustered around the ideal fit line, yielding a mean absolute error of 0.03 V and a Pearson coefficient of 0.974. These practical results confirm that the PINN model generalizes well to unseen diodes and excitation profiles and maintains high accuracy.

TABLE I
COMPARISON OF ML REGRESSION MODELS.

Data	Metrics	Poly	DNN	XGBoost	PINN
Simulation	MAE	0.084	0.059	0.045	0.047
	RMSE	0.143	0.081	0.059	0.060
	ρ	0.908	0.979	0.990	0.989
	Paired t-test	6.2e-6	4.0e-15	3.9e-23	7.6e-20
Measurement	MAE	0.082	0.045	0.036	0.030
	RMSE	0.114	0.053	0.043	0.035
	ρ	0.588	0.931	0.967	0.974
	Paired t-test	5.0e-1	9.6e-3	9.6e-3	5.1e-3

C. Comparison of Regression Models

Table I contrasts four surrogate models across the simulated and measurement data, polynomial regression baseline, DNN, XGBoost [21] and PINN. In the simulation test, XGBoost and PINN achieve comparable accuracy, but in the real measurement test, PINN demonstrates superior generalization, outperforming XGBoost and other two models. The polynomial baseline exhibits poor generalization in the measurement test, highlighting the necessity of advanced ML methods for this nonlinear prediction task. Critically, while purely data-driven models suffer a noticeable performance degradation on real test data, the physical constraint based on energy balance allows the PINN model to prevent overfitting and extrapolate robustly to unseen diodes and waveforms, validating its effectiveness for SPICE surrogate approximations.

V. DISCUSSION AND CONCLUSION

In the proposed framework, the PINN delivers diode evaluations in milliseconds with direct datasheet information, even when no validated SPICE model exists, and replaces iterative simulations or prototyping for optimizing diode selection. Although trained solely on synthetic data, it retains high accuracy on real-world test cases thanks to the embedded physics penalty. Such accuracy and robustness enable rapid component screening and fit naturally into automated design loops, including optimization or design-space exploration.

However, the present study employs a lightweight network and a dataset restricted to a narrow set of waveforms, diode types, and a single circuit topology. Future work will explore richer models, enlarge the validation, extend the PINN to multi-stage topologies by investigating physics constraints at specific nodes, and integrate the surrogate into multi-objective optimization flows [22], [23].

In summary, the proposed framework enables fast and accurate estimation of diode performance using datasheet-level inputs, paving the way for future circuit design optimization and automation. This capability supports low-power, ultra-low-frequency PEHs for body energy harvesting by identifying the optimal diode and accelerating the design of miniaturized rectifier stages. This study offers a concrete path toward realizing intracardiac leadless pacemakers and brings the vision of fully self-powered cardiac devices closer to reality.

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